Method Statement

Ref. #: DCP00/08-0199/0200-A-2025



DonProof FB/FB-S

(High-performance, pre-applied, fully bonded PVC membrane for below-grade waterproofing)

Table of Content

SECTION A: GENERAL COMMENTS	3
General Notes	3
System Products	3
Tools and Equipment	4

SECTION B: APPLICATION	5
Substrate Preparation	5
Separation/Protection Layers below the Membrane	5
Installation of DonProof FB/FB-S Membrane on Horizontal Substrates	6
Installation of DonProof FB/FB-S Membrane on Vertical Substrates	7
Detailing of Overlaps	8
Installation on Removable Formwork	10
Cleaning	11

SECTION C: CAUTIONS	11
Health & Safety	11

SECTION D: APPROVAL AND VARIATIONS	11

Section A: General Comments

General Notes:

The information below is a detailed overview of the application of DCP's **DonProof FB/FB-S** waterproofing membrane and should be read in conjunction with the relevant technical data sheet prior to application. All DCP Products should be applied by experienced specialist contractors.

All the points below assume the correct preparation of the relevant surface.

- Guidelines and recommendations as described in BS 8102:2009 should be followed.
- A suitable water management system should be installed to provide a dry environment until such time as the detailing of critical points, the installation of the membrane, the placement and the curing of the concrete is completed.
- Manufacturer's installation instructions must be followed at all times.
- Concrete must be of a high quality, fully compacted, properly placed and properly cured.
- Care should be taken when using vibrating equipment to prevent damage to the installed waterproofing system.
- The waterproofing system should be installed by fully trained personnel.
- All non-movement construction joints must be detailed using **Hydrostop range** PVC waterstops or alternatively **Hyperseal DP** swellable bar.
- Expansion joints must be detailed with **Hydrostop range** PVC waterstops (internal or external).
- Pipe penetrations must be detailed with **Hyperseal DP** swellable bar.
- Piles must be detailed with **Hyperseal DP** waterstop or injection hose system can be used additionally.
- All site details as well as the concrete mix design must be designed in advance.

System Products:

Separation/protection layers below the Membrane: **Geotex** (weighing no less than 300g/m²).
Waterproofing Membrane: **DonProof FB/FB-S.**





Tools and Equipment:

It is suggested that the following list of equipment are adopted as a minimum requirement

Personal protection	:	*Protective overalls*
	:	*Goggles or a face mask*
	:	*Good quality gloves*
	:	*Safety shoes*
	:	*Safety helmet*
Application equipment	:	*Manual hot-air blower (Fig.1)*
	:	*Utility knife with blade (Fig.2)*
	:	*Pressure roller (Fig.3)*
	:	*Nail gun + Nails (Fig.4)*
	:	*Measuring & marking tools (Fig.5)*





Fig.1: Manual hot-air blower



Fig.2: Utility knife with blade



Fig.3: Pressure roller



Fig.4: Nail Gun + Nails



Fig.5: Measuring & marking tools

Section B: Application

1.0 Substrate Preparation

1.1 Ensure all substrates are solid, clean, and free of debris, sharp edges, or voids greater than 12 mm.

1.2 Lean-mix concrete or compacted sands are suitable base substrates.

1.3 All upstands, corners, penetrations, and interfaces must be properly detailed before installing the waterproofing membrane.

1.4 A 20 mm fillet should be formed at the base of the retaining walls, formwork stop ends, and upstands using a modified cementitious mortar.

1.5 The membrane is designed to bond directly with fresh reinforced concrete; thus, surface preparation must follow EN 1992 design standards.



2.0 Separation/Protection Layers below the Membrane

2.1 A cushion layer of geotextile membrane, weighing no less than 300 g/m², shall be installed directly over the blinding concrete as part of the preparation process.

2.2 Prior to placement, the blinding surface should be thoroughly cleaned to ensure it is free from any loose materials, sharp projections, or debris that could damage the membrane.

2.3 The geotextile will be laid loosely across the blinding concrete and positioned to provide a minimum overlap of 100 mm between adjacent sheets, ensuring full coverage of the underlying blinding concrete.

2.4 If the blinding concrete surface is perfectly smooth, the geotextile layer is not required yet highly recommended.





3.0 Installation of DonProof FB/FB-S Membrane on Horizontal Substrates

3.1 Preparation

3.1.1 All pipe penetrations, service ducts etc. must be detailed prior to placing the waterproofing membrane.

3.1.2 Use a modified cementitious mortar to form a 20 mm fillet at the base of the retaining walls/form-work stop ends and upstands.

3.2 Formwork stop-ends

3.2.1 **DonProof FB/FB-S** membrane should be placed onto the properly prepared surface with the fleece facing towards the concrete to be poured.

3.2.2 Install the first length of **DonProof FB/FB-S** membrane in order that one end is flush with the top of the form-work stop-end, and is allowed extend down over the fillet and out onto the properly prepared substrate by at least 300 mm.

3.2.3 Secure the membrane using nail-fixings at 500 mm centers along the top of the form-work stop-end.

3.3 Retaining walls

3.3.1 Install the first length of **DonProof FB/FB-S** membrane in order that one end extends up the retaining wall at least 500mm, and that the other end extends down over the fillet and out onto the properly prepared substrate.

3.3.2 Secure the membrane to the retaining wall using nail-fixings at 500 mm centers along the membrane.

3.3.3 Where the reinforced concrete slab will be in greater than 1meter in thickness, install a second length of membrane on the retaining wall in order that the installed membrane on the wall will be at least 200 mm higher than the top of the poured ground slab.

3.3.4 All overlaps of the membranes should be a minimum of 50 mm and must be detailed by a heat welding system, and compressed using a wooden roller.



4.0 Installation of DonProof FB/FB-S Membrane on Vertical Substrates

4.1 All substrates onto which **DonProof FB/FB-S** membrane is to be installed should be flat and free from sharp objects and voids that are greater than 12 mm.

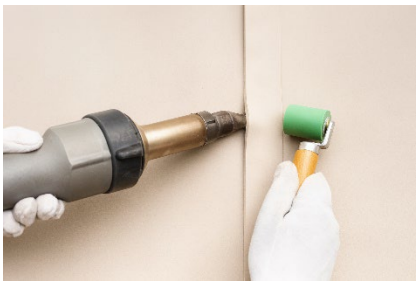
4.2 All pipe penetrations, service ducts etc. must be detailed prior to placing the waterproofing membrane.

4.3 **DonProof FB/FB-S** membrane should be placed onto the properly prepared surface with the fleece facing towards the concrete to be poured.

4.4 **DonProof FB/FB-S** membrane should be installed on properly prepared vertical surfaces by cutting the membrane into the required lengths, securing the membrane at the top using nail-fixings, and allowing it to extend down over the previously installed under-slab membrane by at least 100 mm.

4.5 If required additional nail-fixings may be used placed at 500 mm centers along the clear selvedge where they will be covered by subsequent rolls, should any nail-fixings be exposed these should be covered with a patch of the membrane bonded by a heat welding.





4.6 Compress the patch area using a wooden-roller to ensure complete adhesion.

4.7 Using a clean dry cloth, clean the face of the selvedge on the installed membrane.

4.8 Position the subsequent membrane in order the clean edge-selvedge on the installed membrane will be overlapped by a minimum of 50 mm.

4.9 All overlaps must be sealed by a heat welding system. Use a wooden-roller to compress the overlaps to ensure complete adhesion.



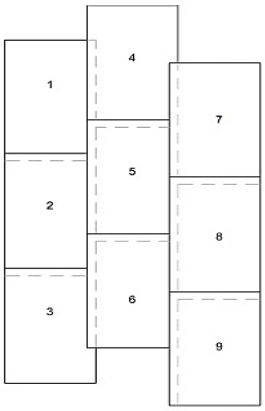
5.0 Detailing of Overlaps

5.1 **Welding side-lap overlaps:**

5.1.1 Using a clean dry cloth, clean the face of the selvedge on the first roll.

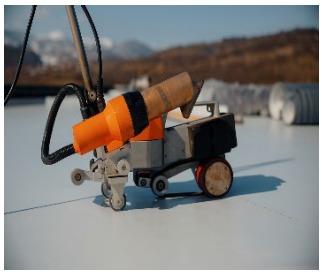
5.1.2 Position the subsequent roll in order that the edge-selvedge on the previous roll will be overlapped by a minimum of 50 mm.





5.1.3 All overlaps must be detailed by a heat welding system, and compressed using a wooden roller.







5.2 **Welding roll end overlaps:**

5.2.1 The ends of the membrane should be staggered by a minimum of 300 mm, to achieve this, simply butt the ends of the subsequent roll tightly together, ensuring there are no gaps greater than 12 mm between the roll ends.

5.2.2 Cut a strip of **DonProof FB/FB-S** membrane that is 150 mm wide, wipe the back of the membrane with a clean cloth, place the strip of membrane over the center of the butt-joint and weld it by a heat welding system.

5.2.3 Use a wooden-roller to compress the overlaps to ensure complete adhesion





5.3 **Welding reinforcing strips:**

5.3.1 Reinforcing strips of **DonProof FB/FB-S** membrane should be used at both internal and external corners, these reinforcing strips may be formed using a strip of **DonProof FB/FB-S** membrane and bonded using the same method as the overlaps.





5.4 **Placement of reinforcement:**

5.4.1 The steel reinforcement should be placed as soon as possible after the membrane is installed.

5.4.2 Do not use reinforcement carriers which have sharp ends.

5.4.3 Flat cement-based carriers should be used.

5.4.4 Where possible place the carriers along the overlaps.

5.5 **Delay in placement of the reinforcement/concrete:**

5.5.1 Where there will be a delay in the placing of the reinforcement or the concrete, extend the membrane by at least 1 m past the termination point of the reinforcement/concrete, fold-back the extended membrane and cover with plastic to ensure it is kept free from damage, clean and dry.





5.6 **Cleaning the membrane:**

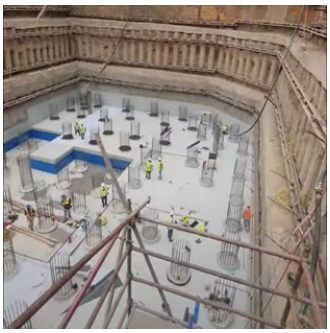
5.6.1 If required, the **DonProof FB/FB-S** membrane may be brushed clean using clean cold water.

5.6.2 The surface of the installed membrane should be free from standing water before placing the concrete.

5.7 **Damages:**

5.7.1 The membrane should be inspected for damages prior to the placement of the concrete.

5.7.2 In the unlikely event of damages being found, these must be repaired using an oversize patch of **DonProof FB/FB-S** membrane, which must be detailed by a heat welding system, and compressed using a wooden roller.





6.0 Installation on Removable Formwork

6.1 **Installation of DonProof FB/FB-S membrane:**

6.1.1 The formwork should be free from voids or damages that are greater than 12 mm.

6.1.2 Using a sharp blade cut the membrane to the required length.

6.1.3 The membrane should be placed against the formwork with the fleece facing towards you/towards the later placed fresh concrete.

6.1.4 Place the first strip of membrane against the formwork. Secure the membrane to the top of the formwork using nail-fixings, fixed at 300 mm centers, additional nail-fixings placed at 500 mm centers along the edge of the membrane may be used if required. Ensure the membrane is laid flat against the form-work.

6.1.5 Using a clean dry cloth wipe the face of the selvedge on the installed membrane.

6.1.6 Position the subsequent length of membrane, ensuring the edge-selvedge on the previous installed membrane will be overlapped by a minimum of 50 mm, the overlaps must be sealed by a heat welding system and compressed using a wooden roller.

6.1.7 The membrane should be inspected for damages prior to installing the formwork.

6.1.8 In the unlikely event of damages been found these must be repaired using an oversized patch of **DonProof FB/FB-S** membrane bonded by a heat welding system and compressed using a wooden roller.

6.1.9 The fresh concrete must achieve a minimum compressive strength of 10 K/mm² before the formwork is removed.

6.1.10 All tie-holes should be filled with waterproofing Mortar and an oversize patch of **DonProof FB/FB-S** membrane bonded by a heat welding system and compressed using a wooden roller.

6.1.11 Should the ends of any nail-fixings be protruding through the membrane, these should be cut off and the area covered with a patch of **DonProof FB/FB-S** membrane bonded by a heat welding system and compressed using a wooden roller.

6.2 **Backfilling:**

6.2.1 The Walls should be inspected for damages prior to backfilling, should there be any damages, honeycombing etc., these areas should be repaired and detailed with an oversize patch of **DonProof FB/FB-S** membrane bonded by a heat welding system and compressed using a wooden roller.

6.2.2 There is no requirement for protection board with **DonProof FB/FB-S** membrane, provided care is taken when backfilling.

6.2.3 Backfill should be clean filling that is free from sharp objects or construction debris. Do not use stones or filling that are larger than 50 mm.

6.2.4 Backfill should be compacted every 300 mm as defined by ASTM 1557.

6.2.5 Care should be taken to prevent damage to the waterproofing membrane when back-filling.

7.0 Cleaning

7.1 Tools and equipment can be cleaned with water.

Section C: Cautions

Health and safety

- Apply in well-ventilated areas. In closed areas use force ventilation and wear approved respirators when necessary.
- **DonProof FB/FB-S** should not come into contact with skin and eyes. In case of contact with eyes wash immediately with plenty of water and seek medical advice promptly.
- Vapor and atomized liquids are harmful.

Fire:

DonProof FB/FB-S is nonflammable.

For further information on refer to the Material Safety Data Sheet.

Section D: Approval and Variations

This method statement is offered by DCP as a 'standard proposal' for the application of **DonProof FB/FB-S**. It remains the responsibility of the Engineer to determine the correct method for any given application. Where alternative methods are to be used, these must be submitted to DCP for approval, in writing, prior to commencement of any work. DCP will not accept responsibility or liability for variations to the above method statement under any other condition.